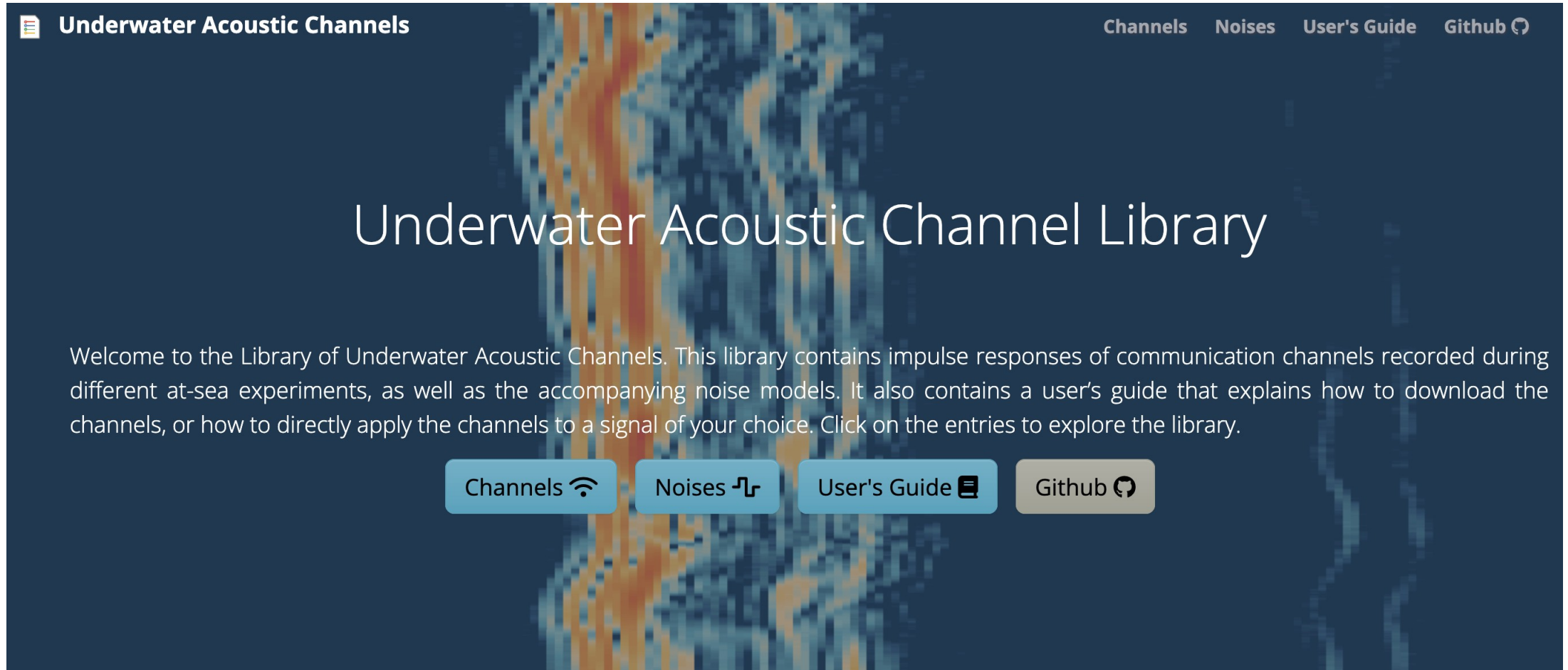


Underwater Acoustic Channel Library

– A Hands-On Tutorial –



The screenshot shows the homepage of the 'Underwater Acoustic Channels' website. The background is a dark blue with a large, abstract, orange and yellow wave-like pattern. At the top left, there is a logo consisting of three horizontal bars of increasing height, followed by the text 'Underwater Acoustic Channels'. At the top right, there are four navigation links: 'Channels', 'Noises', 'User's Guide', and 'Github', each with a small icon. In the center, the title 'Underwater Acoustic Channel Library' is displayed in a large, white, sans-serif font. Below the title, a paragraph of text reads: 'Welcome to the Library of Underwater Acoustic Channels. This library contains impulse responses of communication channels recorded during different at-sea experiments, as well as the accompanying noise models. It also contains a user's guide that explains how to download the channels, or how to directly apply the channels to a signal of your choice. Click on the entries to explore the library.' At the bottom, there are four buttons: 'Channels' with a signal icon, 'Noises' with a waveform icon, 'User's Guide' with a document icon, and 'Github' with a GitHub logo icon. The buttons are light blue with rounded corners.

Underwater Acoustic Channels

Channels Noises User's Guide Github

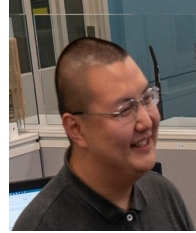
Underwater Acoustic Channel Library

Welcome to the Library of Underwater Acoustic Channels. This library contains impulse responses of communication channels recorded during different at-sea experiments, as well as the accompanying noise models. It also contains a user's guide that explains how to download the channels, or how to directly apply the channels to a signal of your choice. Click on the entries to explore the library.

Channels Noises User's Guide Github

Part 1: Background

Zhengnan
Li



Paul van
Walree



Jim
Preisig



Milica
Stojanovic



Andy
Singer

Mandar
Chitre

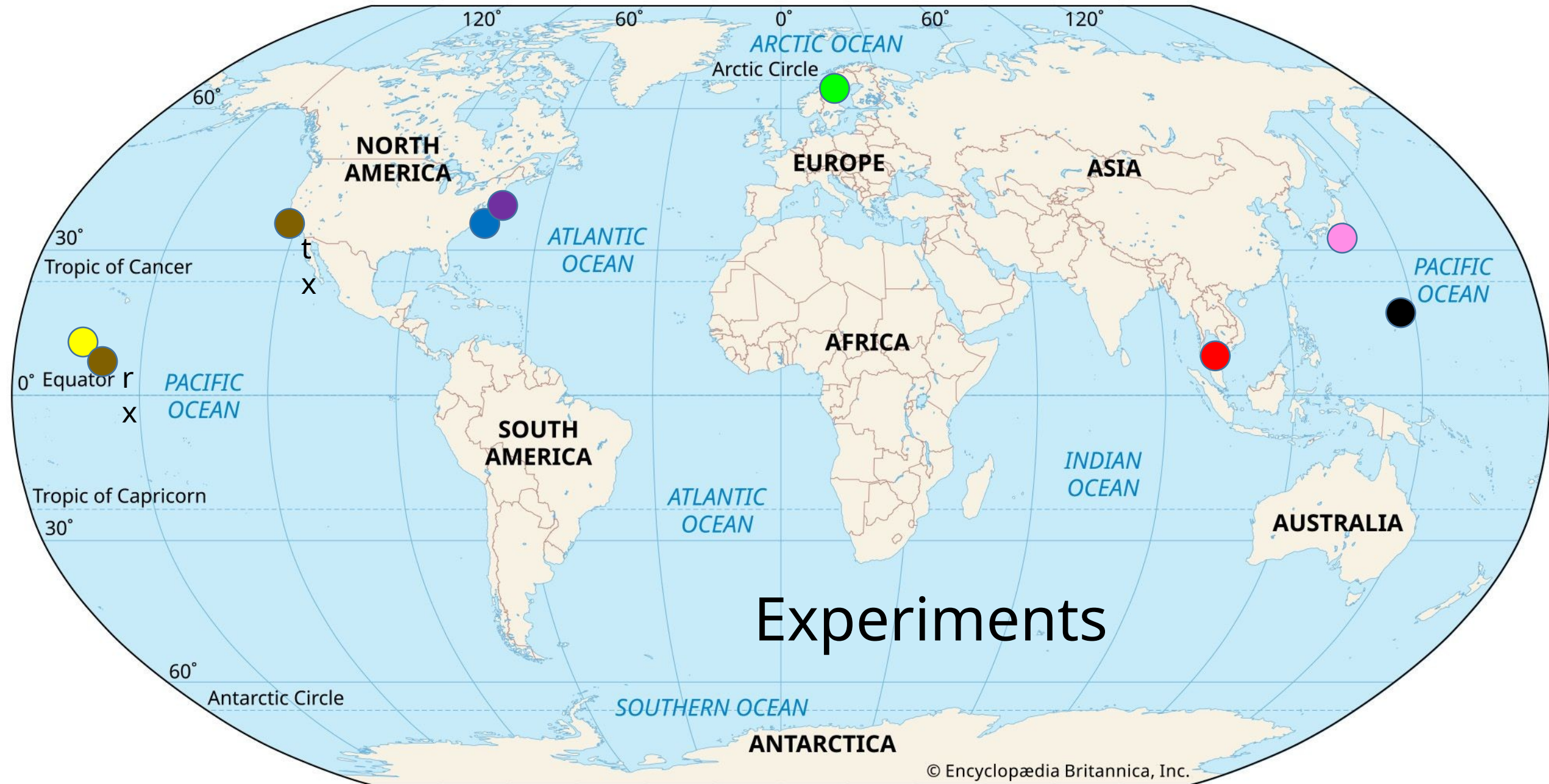


Diego
Cuji



Why do we need a library of channels?

- Design of any communication system begins with some knowledge about the propagation channel.
- Radio channels have **standardized** models; acoustic channels do not.
- Standardized models provide a **common** ground for design, testing, and performance comparison of candidate communication algorithms and protocols. They are universally available, free of charge.
- In the absence of standardized models, design of acoustic communication systems must rely on experiments. Experiments are costly and time consuming. They are available only to a select few (the equipment *and* the data).



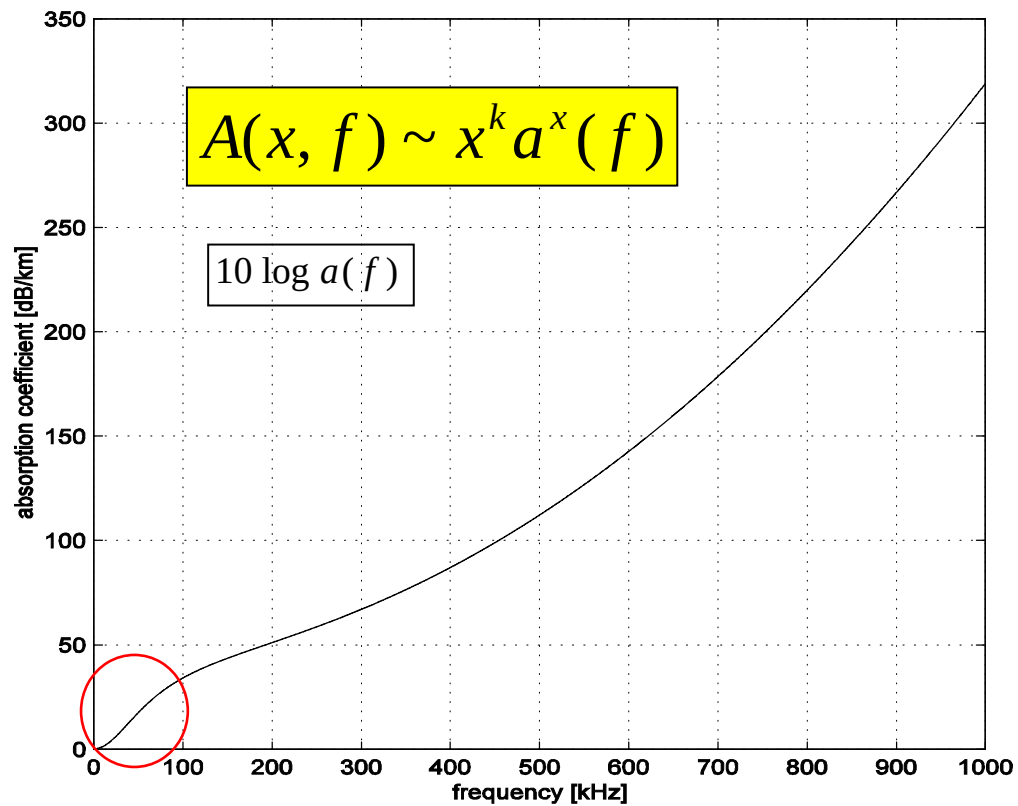
The library

- The library is universally available online.
- It contains
 - a variety of channel models extracted from at-sea recordings (short/long range, mobile/fixed platforms, geographically diverse locations).
 - a simple-to-use interface for passing a custom-designed signal through a channel
- The channels capture
 - multipath (delay spreading/frequency selectivity)
 - random variations (Doppler spreading)
 - Doppler shifting due to motion (time compression/dilation)
 - noise

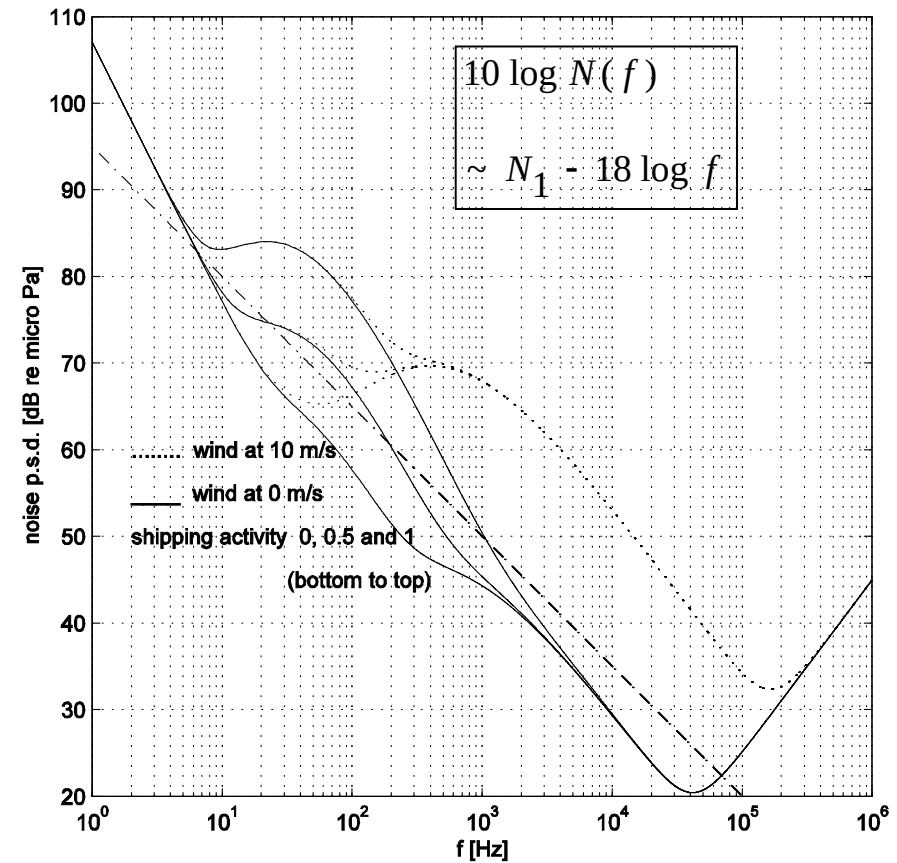
Basics of Acoustic Propagation: Why Radio Models Don't Work

Path loss and noise

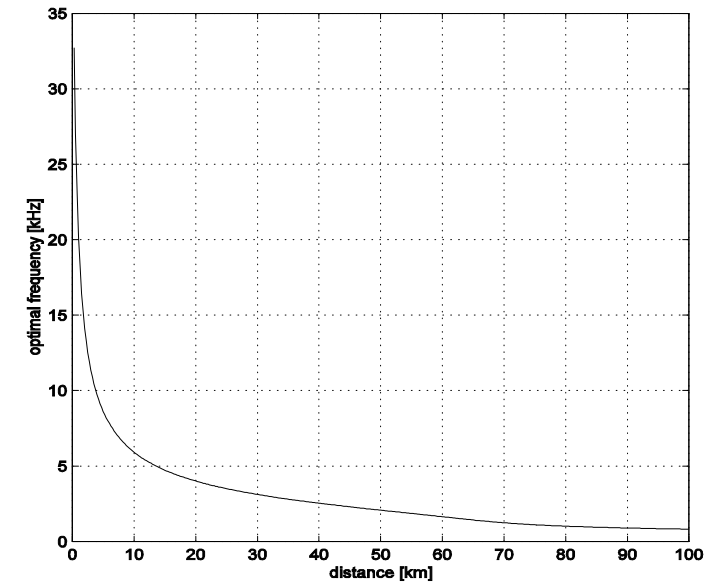
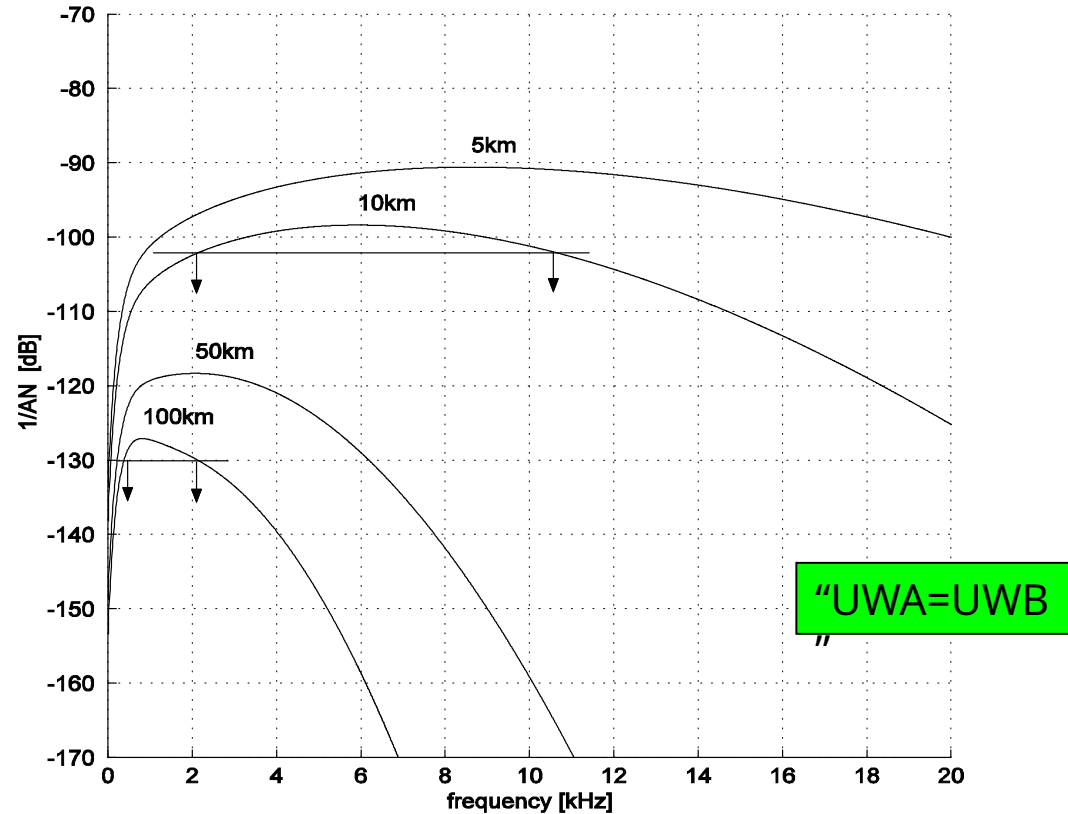
Path
loss



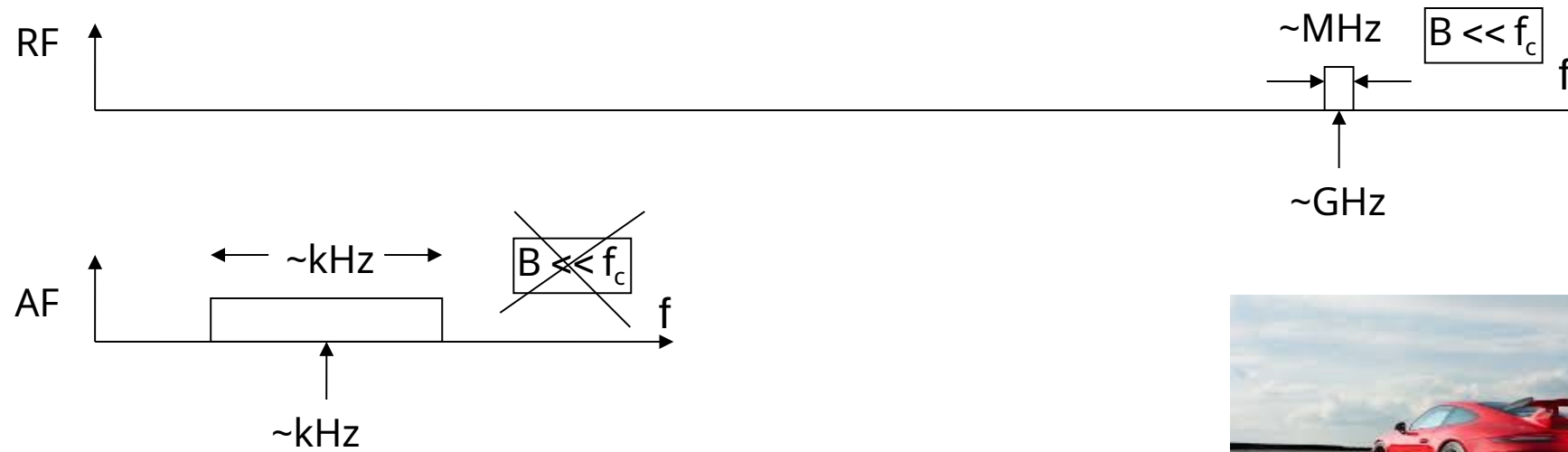
Ambient
noise



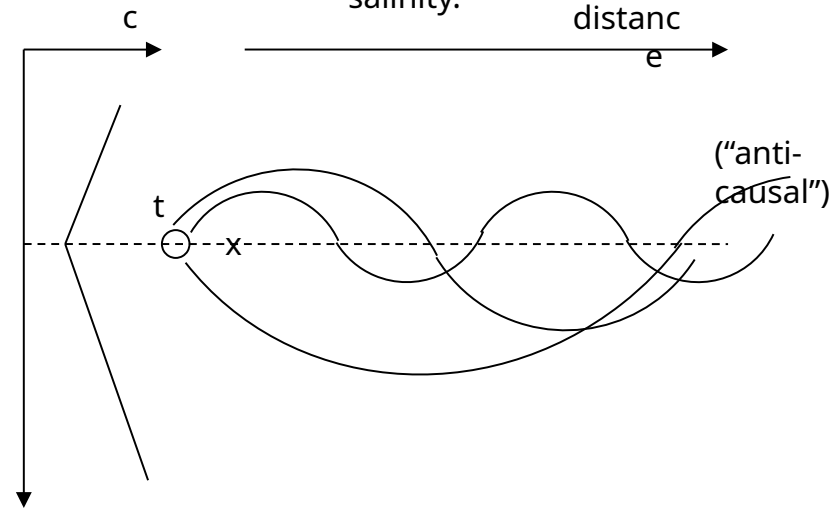
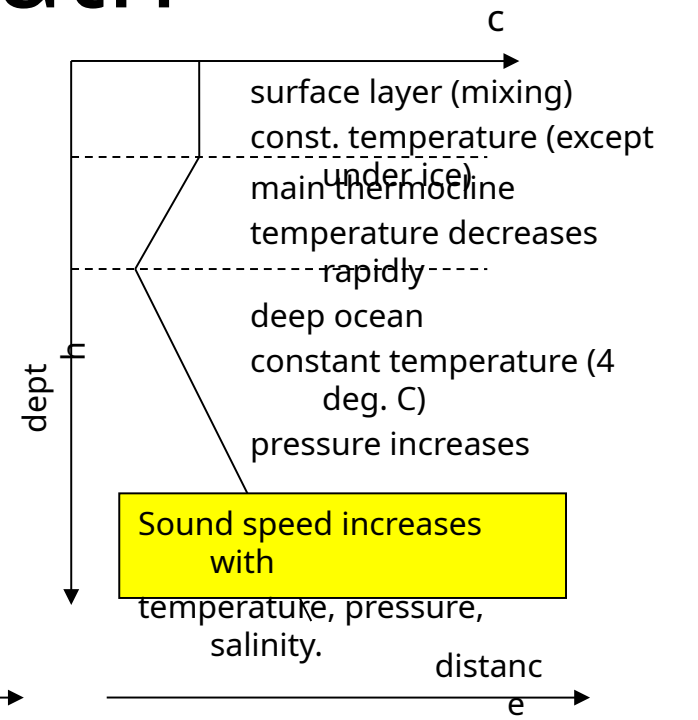
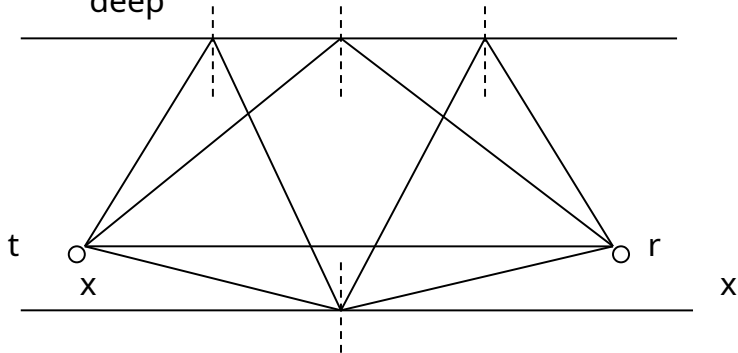
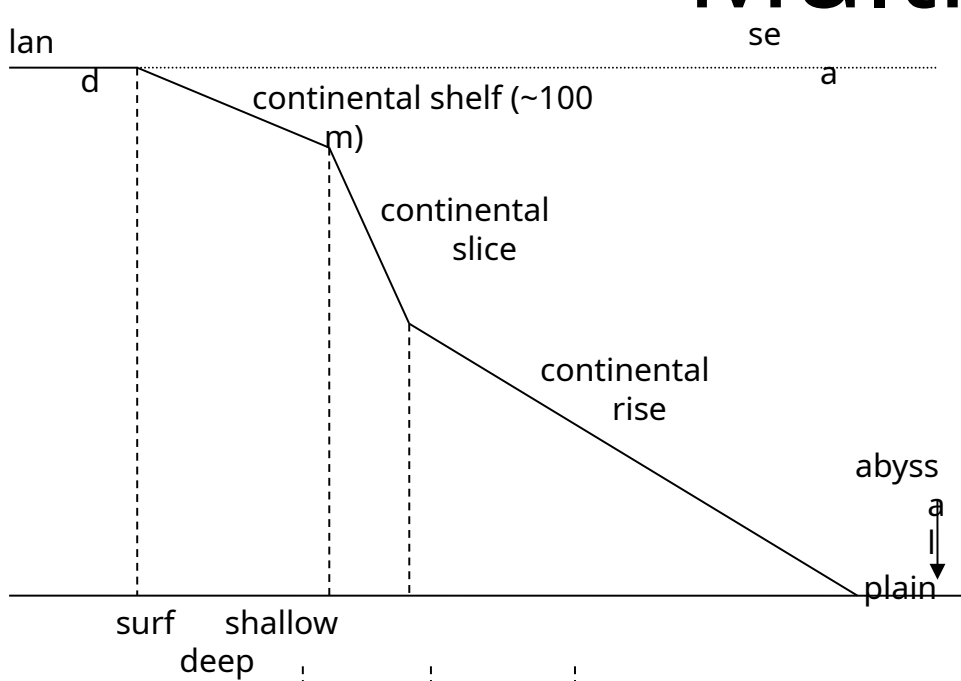
Bandwidth-distance dependence



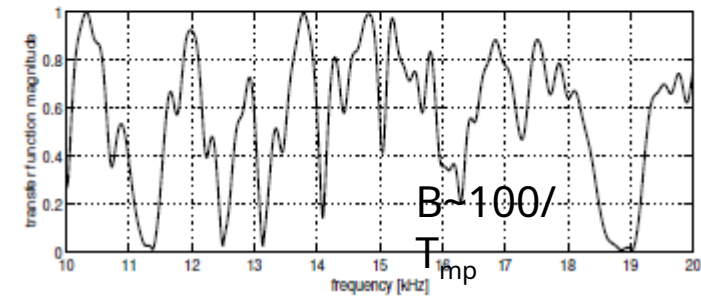
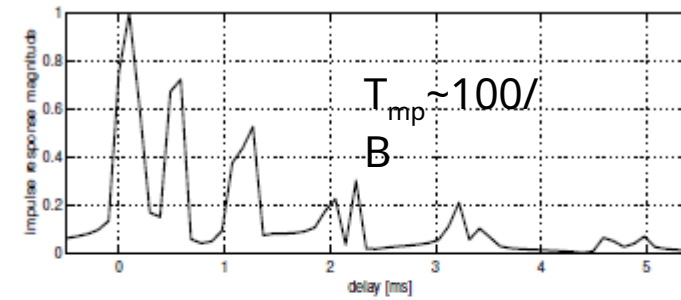
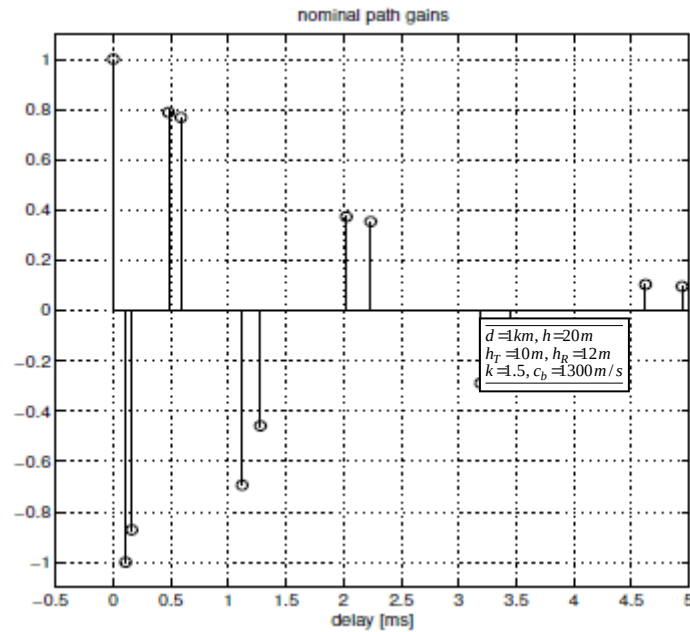
AF vs. RF



Multipath

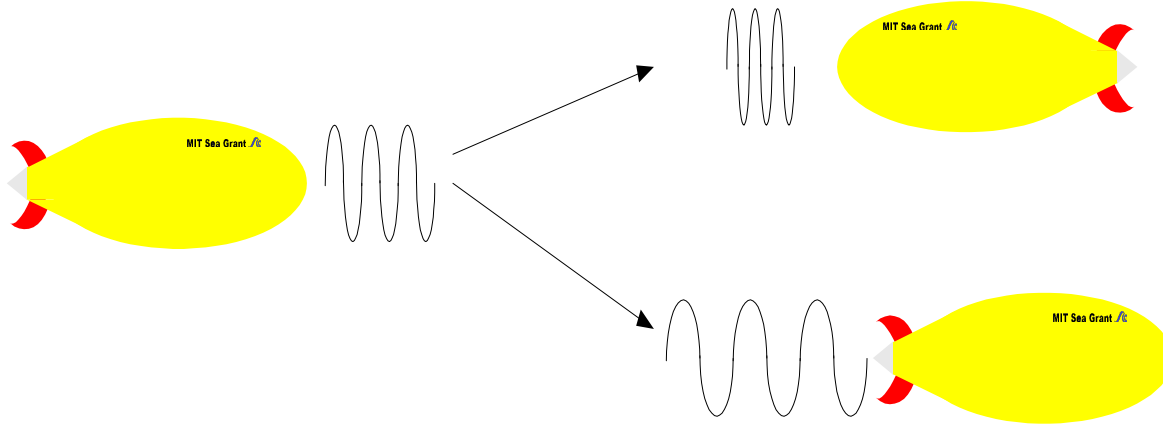


Frequency selectivity

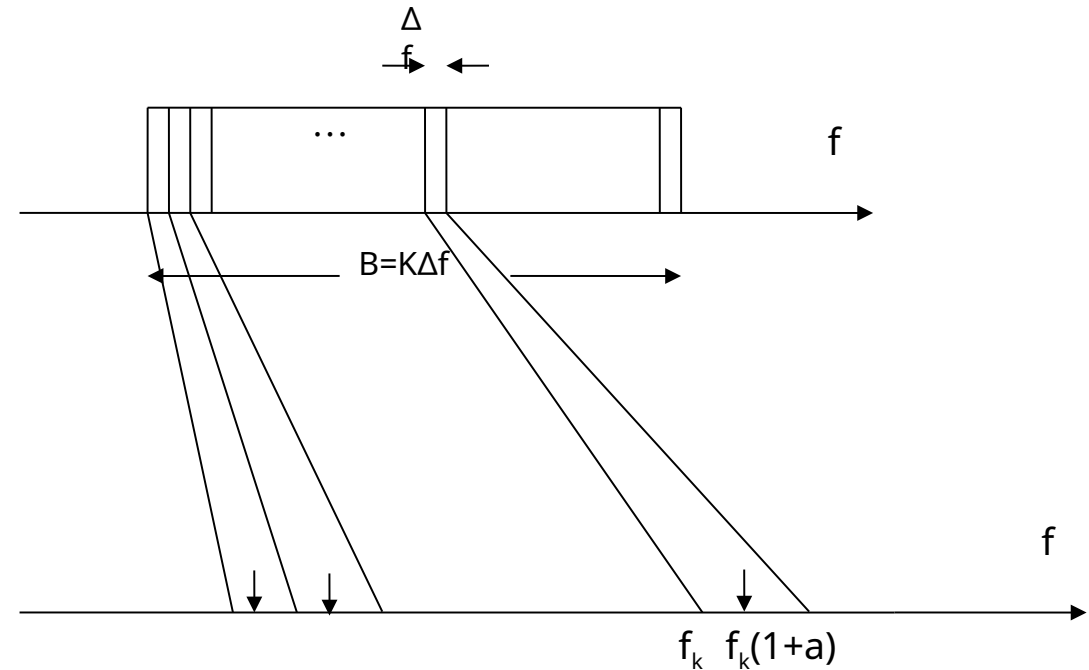


Channel must be equalized
(single-carrier or multi-carrier).

Doppler shifting



$$c=1500 \text{ m/s}, v \sim \text{m/s}$$



$$a'=v/c \sim 10$$

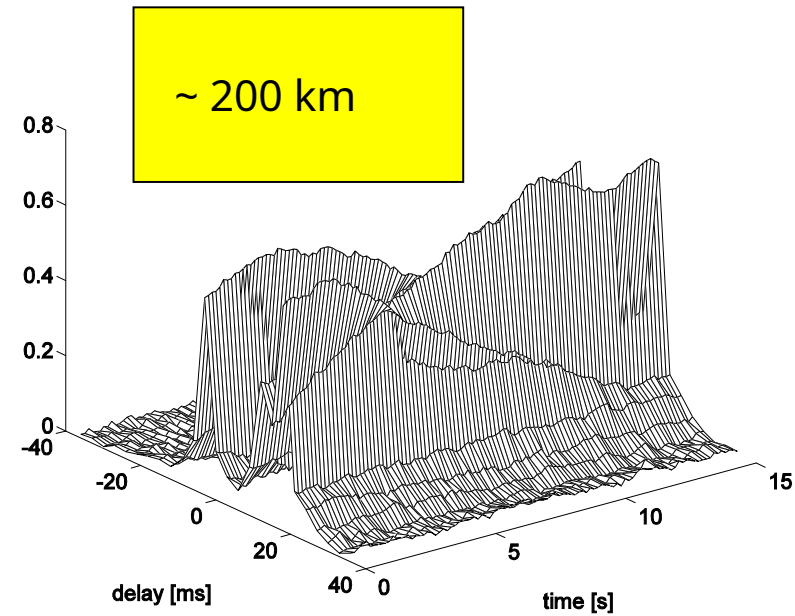
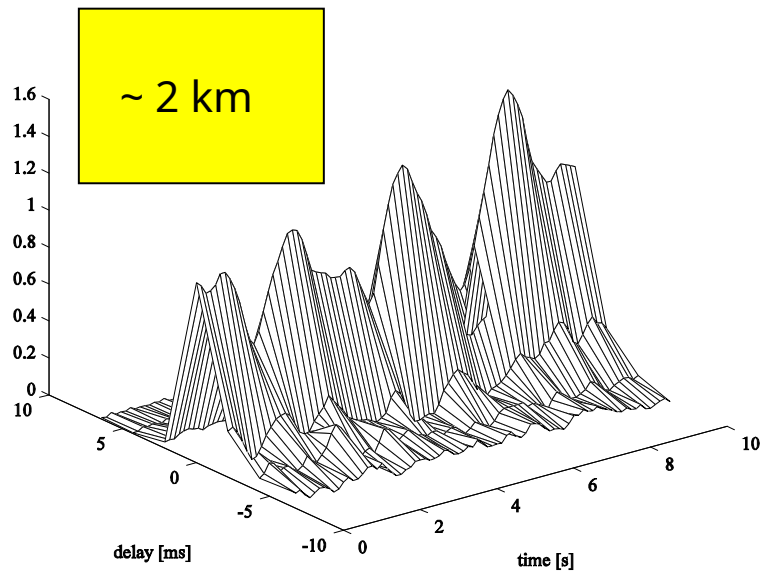
(compare to 10^{-7} @ 65 mph RF)

$$\begin{matrix} t & t+a't \\ f & f+a'f \end{matrix}$$

time
dilation/compression
frequency offset

Synchronization is essential.

Random variations (fading)



There are no standardized statistical models for spatial/temporal channel variability in "typical" locations.

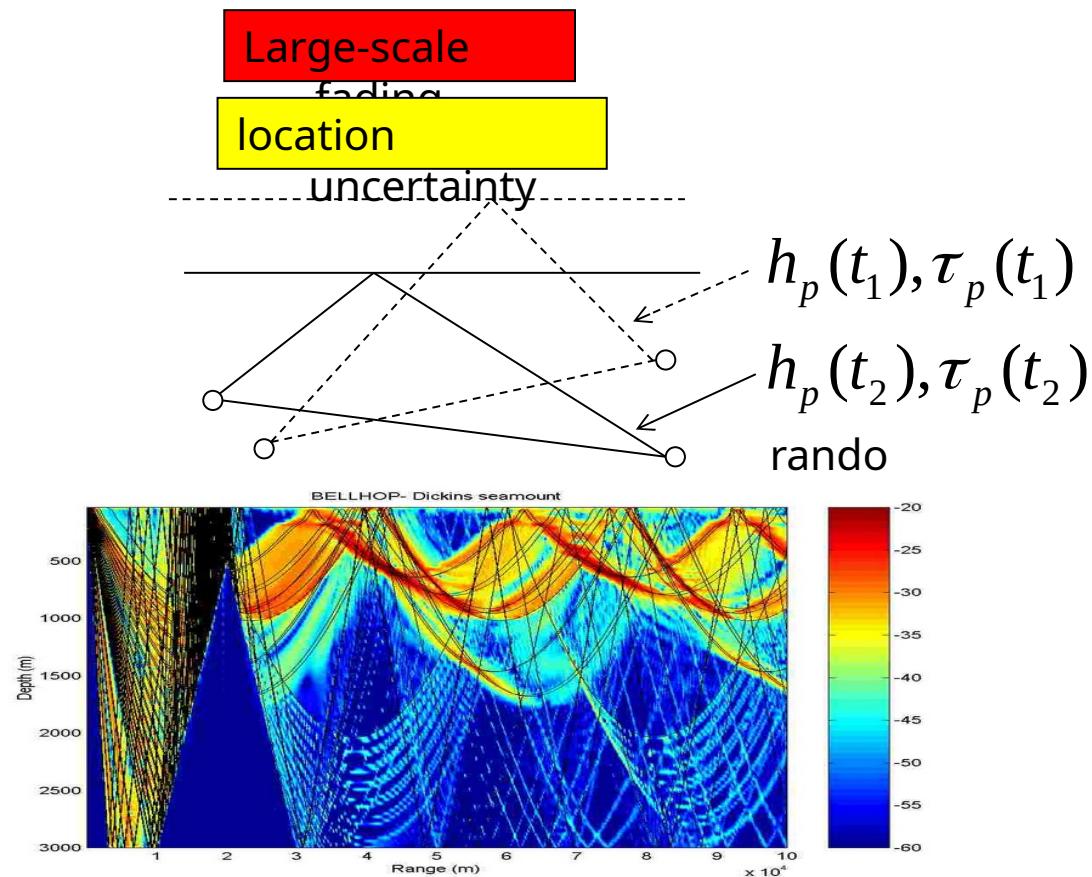
Channel extraction: Theoretical library foundation

Two approaches to channel extraction

1. As recorded:
channel is estimated from experimentally recorded data
(e.g, van Walree et al.. "The Watermark...", IEEE JoE 2017)
2. Propagation modeling:
deterministic nominal conditions + statistical modeling of
random effects
(e.g., Qarabaqi and Stojanovic, "Statistical...", IEEE JoE 2013)

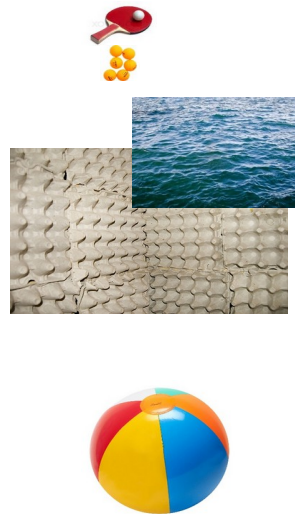
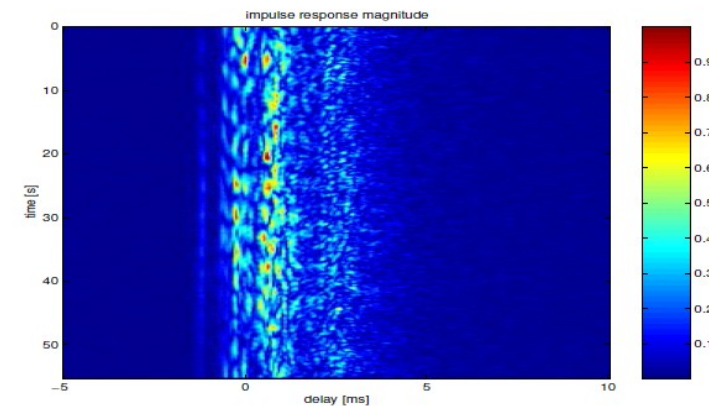
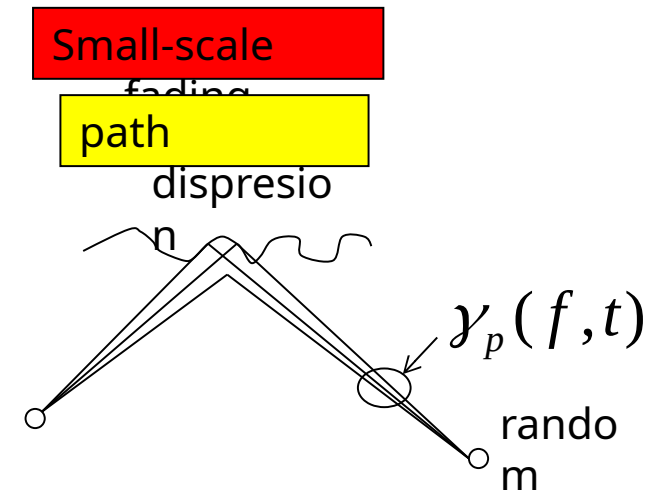
Propagation modeling

deterministic + random effects



$$H(f, t) = \sum_p h_p(t) \gamma_p(f, t) e^{-j2\pi f \tau_p(t)}$$

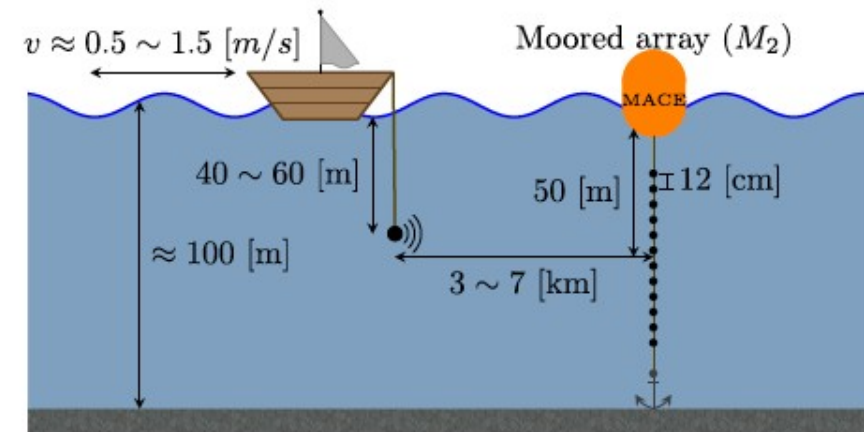
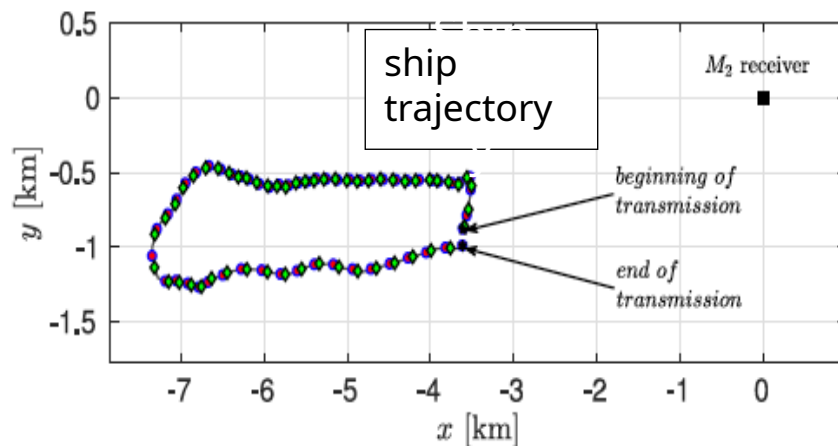
$$\tau_p(t) = \tau_p(t_n) - a_p(t - t_n), t \in T_n$$





Example: Blue channel

- Location: Rhode Island $40^{\circ}21'N$, $70^{\circ}47'W$
- Date recorded: June 2010
- Band: 10.5-15.5 kHz
- Distance: 3-7 km
- Tx/Rx/water depth [m]: 30-60/50/100
- Mobility: varying, up to 1.5 m/s
- Receiving array: 12 elements, 12 cm spacing



Signal Processing

- During each experiment, a probe signal is transmitted.
- The received signal is recorded for off-line processing.
- The probe signal is a pseudo-random sequences modulated onto a carrier.
- Channels are extracted from experimental recordings by applying an adaptive algorithm for **channel estimation, phase and delay tracking**.
- Channel responses are stored in matrices. Each row of the channel matrix contains the impulse response of the channel estimated at a given moment in time. The evolution of the channel response over time is captured across the rows.
- The channel matrices are compressed and stored in the library.
- Motion-induced Doppler shifting (delay drifting) is decoupled from the rest of the channel and stored separately.
- If there is an array, each element is treated separately (any delay difference is preserved).

Basic channel estimation with phase tracking



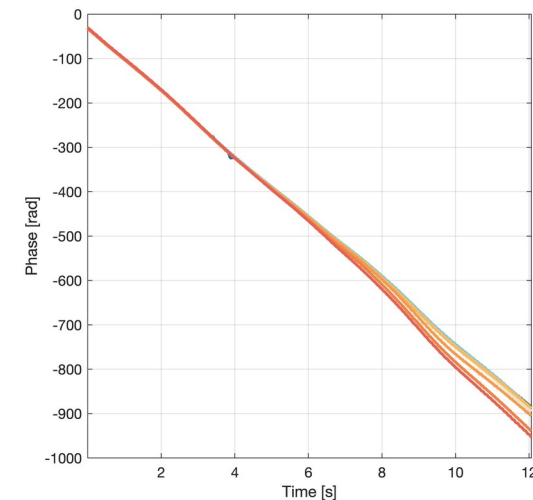
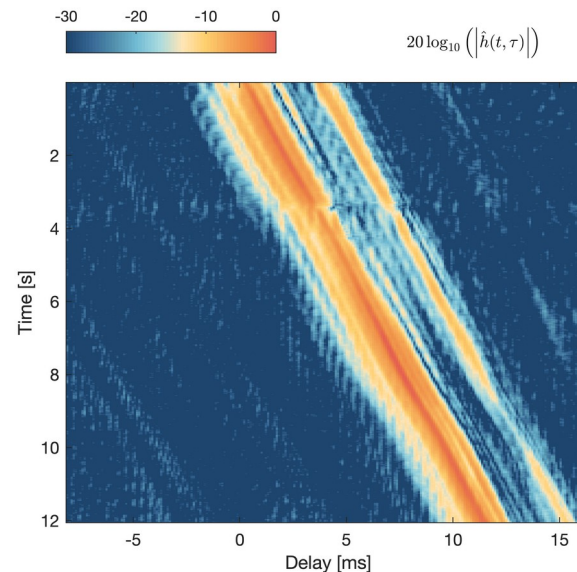
$$v(nT_s) = \mathbf{h}^T[n] \mathbf{x}(n) e^{j\hat{\theta}(nT_s)} + w(nT_s)$$

$$e(nT_s) = v(nT_s) e^{-j\hat{\theta}(nT_s)} - \hat{\mathbf{h}}^T[n] \mathbf{x}(n)$$

$$\psi(nT_s) = -\text{Im}\{\hat{\mathbf{h}}^T[n] \mathbf{x}(n) e^{*}(nT_s)\}$$

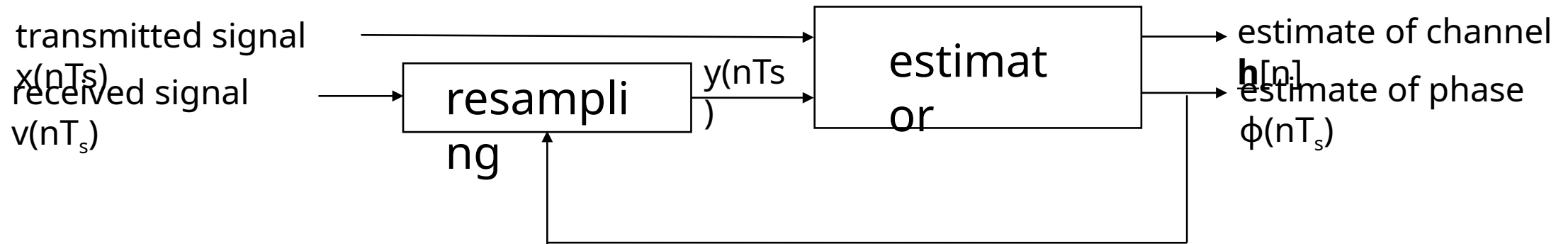
$$\hat{\mathbf{h}}^*[n+1] = \hat{\mathbf{h}}^*[n] + \mathcal{A}[\mathbf{x}(n), e(nT_s)]$$

$$\hat{\theta}(nT_s + T_s) = \hat{\theta}(nT_s) + \text{PLL}[\psi(nT_s)]$$



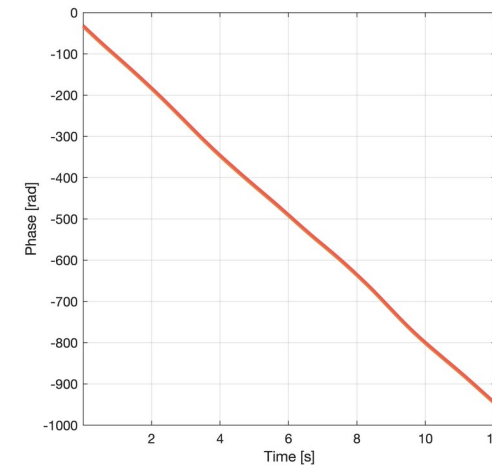
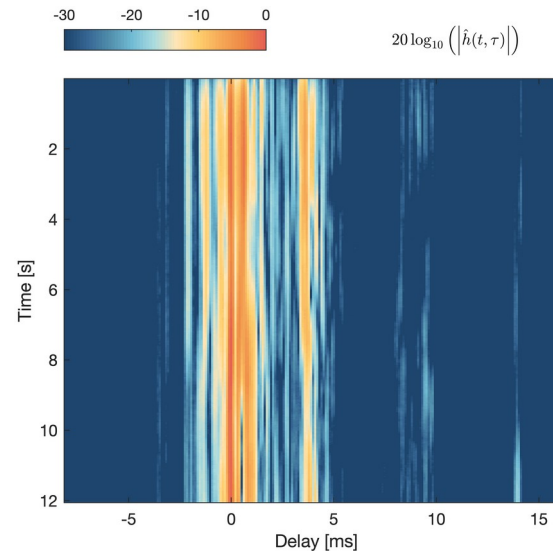
Delay drifting: Capturing a longer time would require a larger delay window which would in turn require an undesirable adjustment of tracking parameters (slower tracking).

Channel estimation with phase and delay tracking



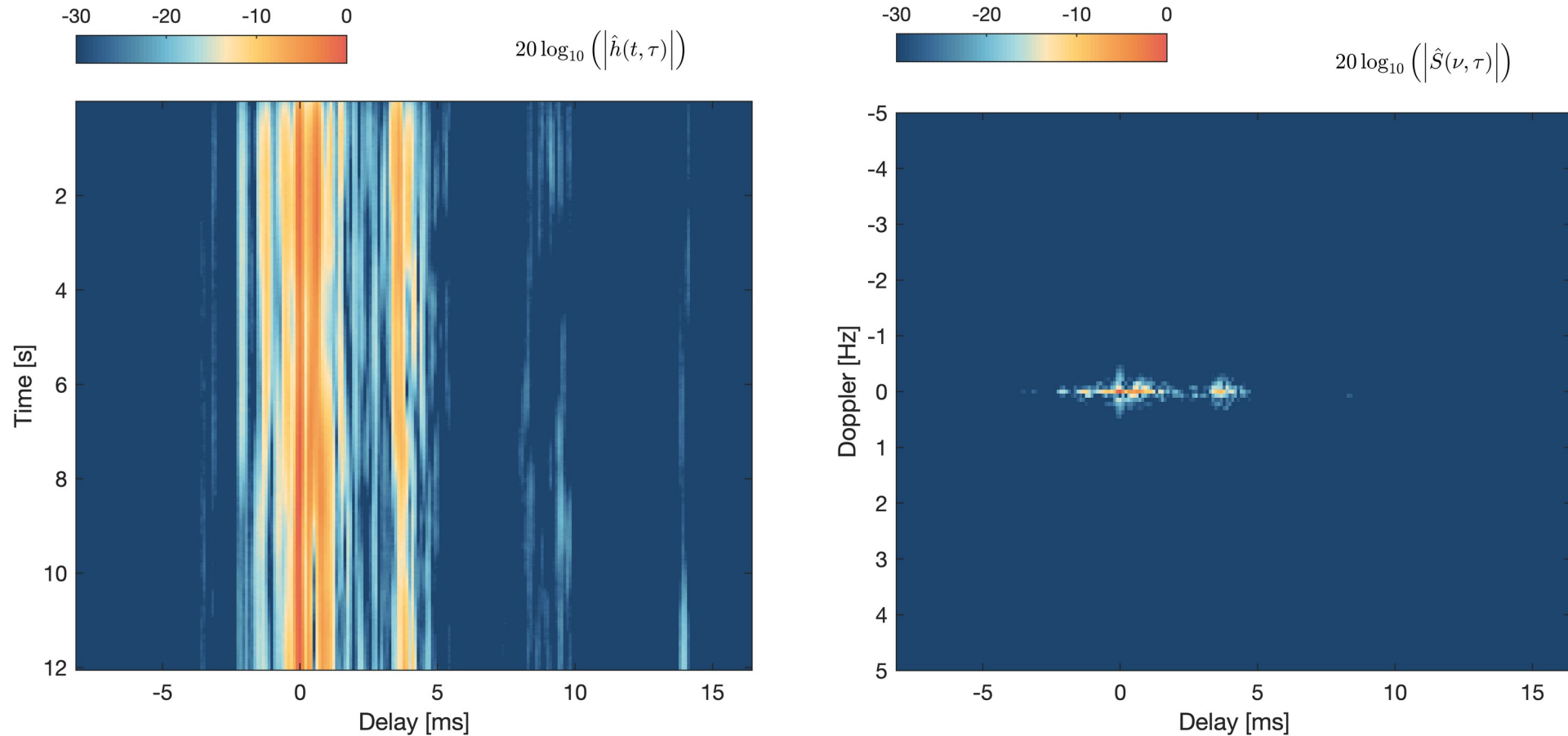
$$v(nT_s) \rightarrow v(nT_s - \epsilon(nT_s))$$

$$\epsilon(nT_s) = \frac{\hat{\phi}(nT_s)}{2\pi f_c}$$



The channel matrix is “straightened.” It can now be efficiently compressed for storage. The delay is stored uncompressed

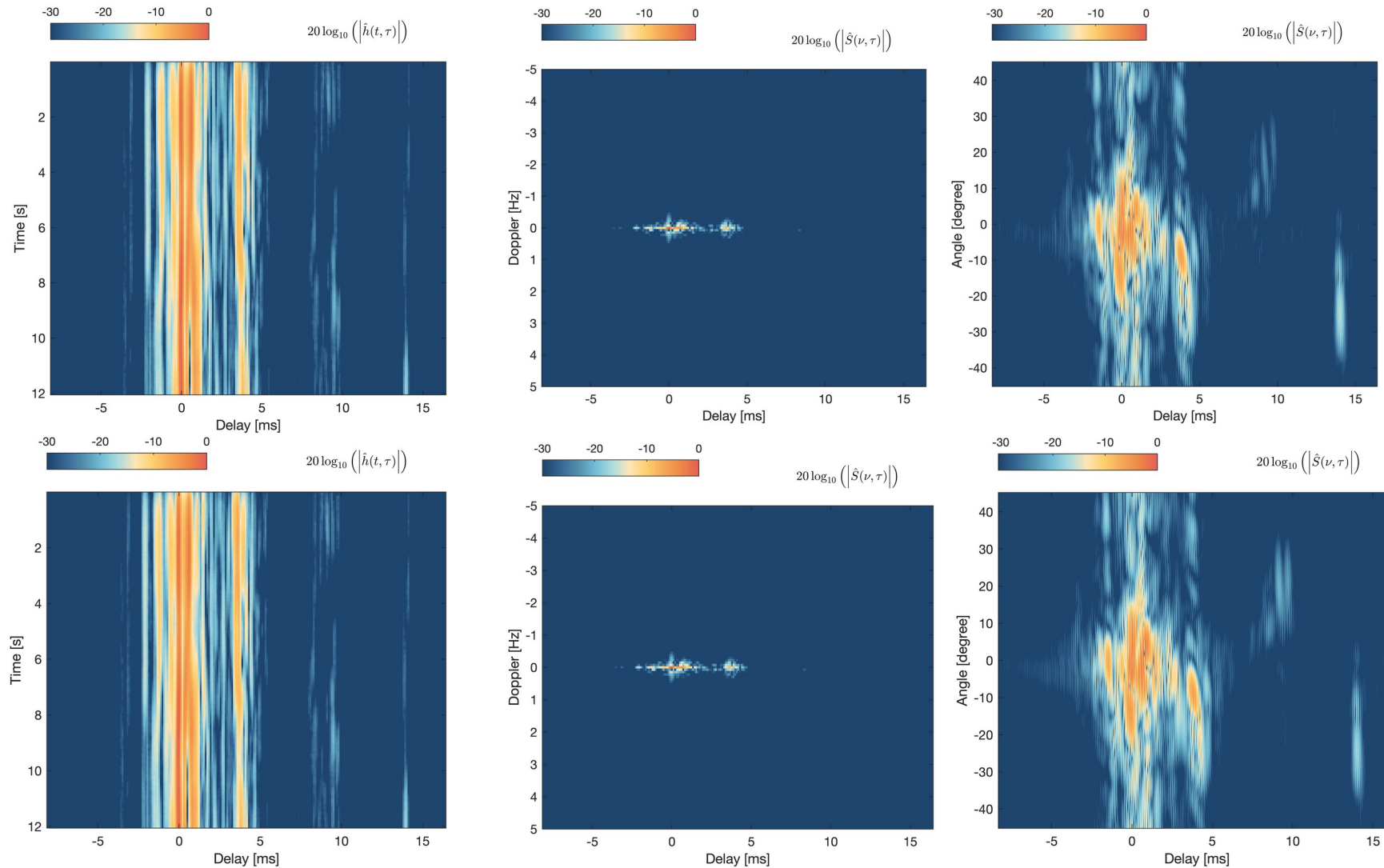
Compression



Doppler spread 1 Hz $\rightarrow$ time compression possible down to 2 samples/second, i.e. by a factor of 5000

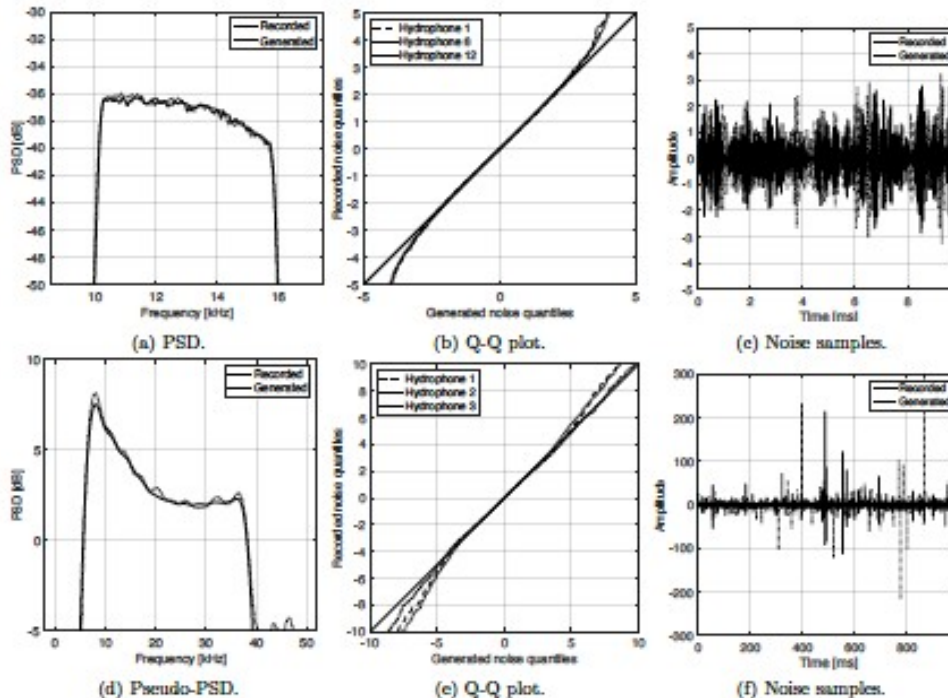
Note: delay "straightening" was a necessary first step.

Reconstruction



Noise

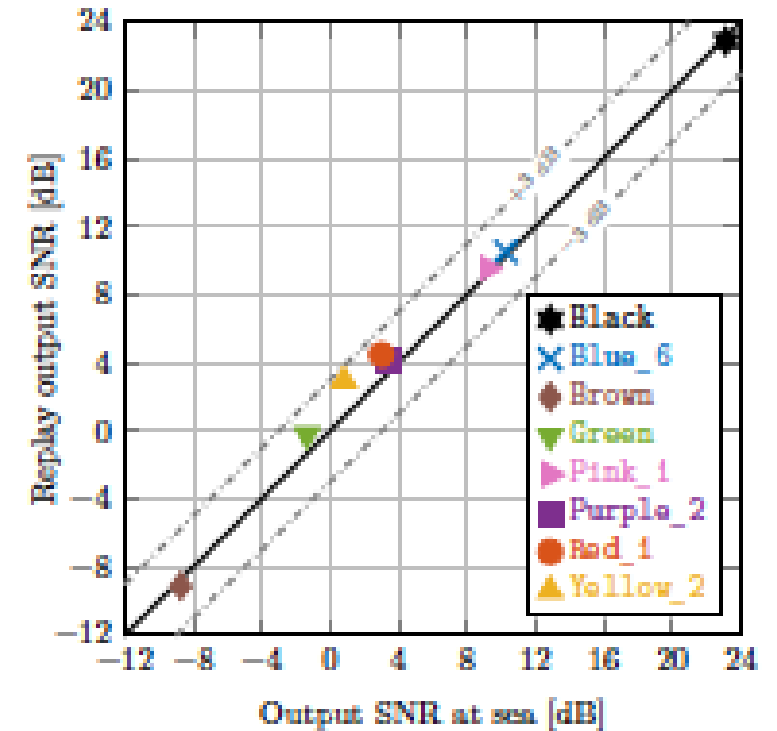
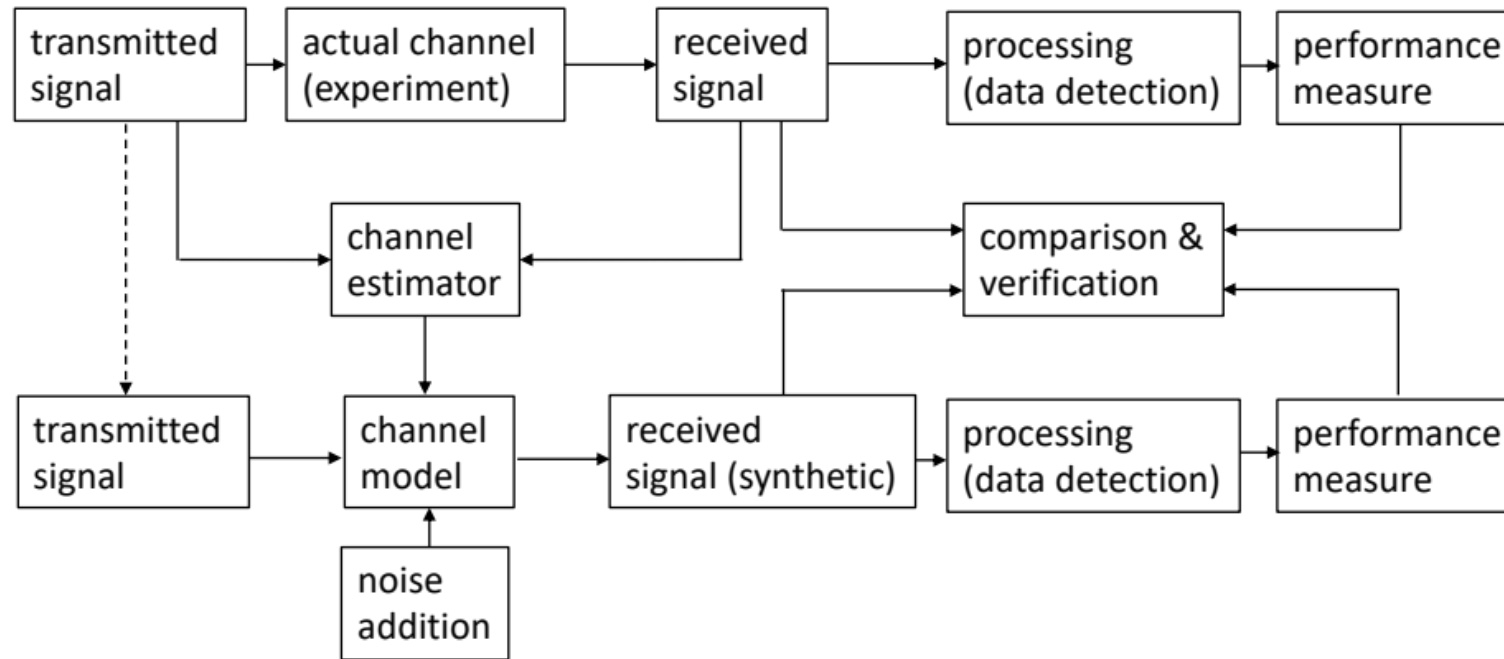
- Noise recorded during each experiment is used to extract statistical properties, which are then used to generate synthetic noise.
- Statistical properties include the probability distribution (pdf) and power spectrum (psd). Both temporal and spatial correlations are modeled.



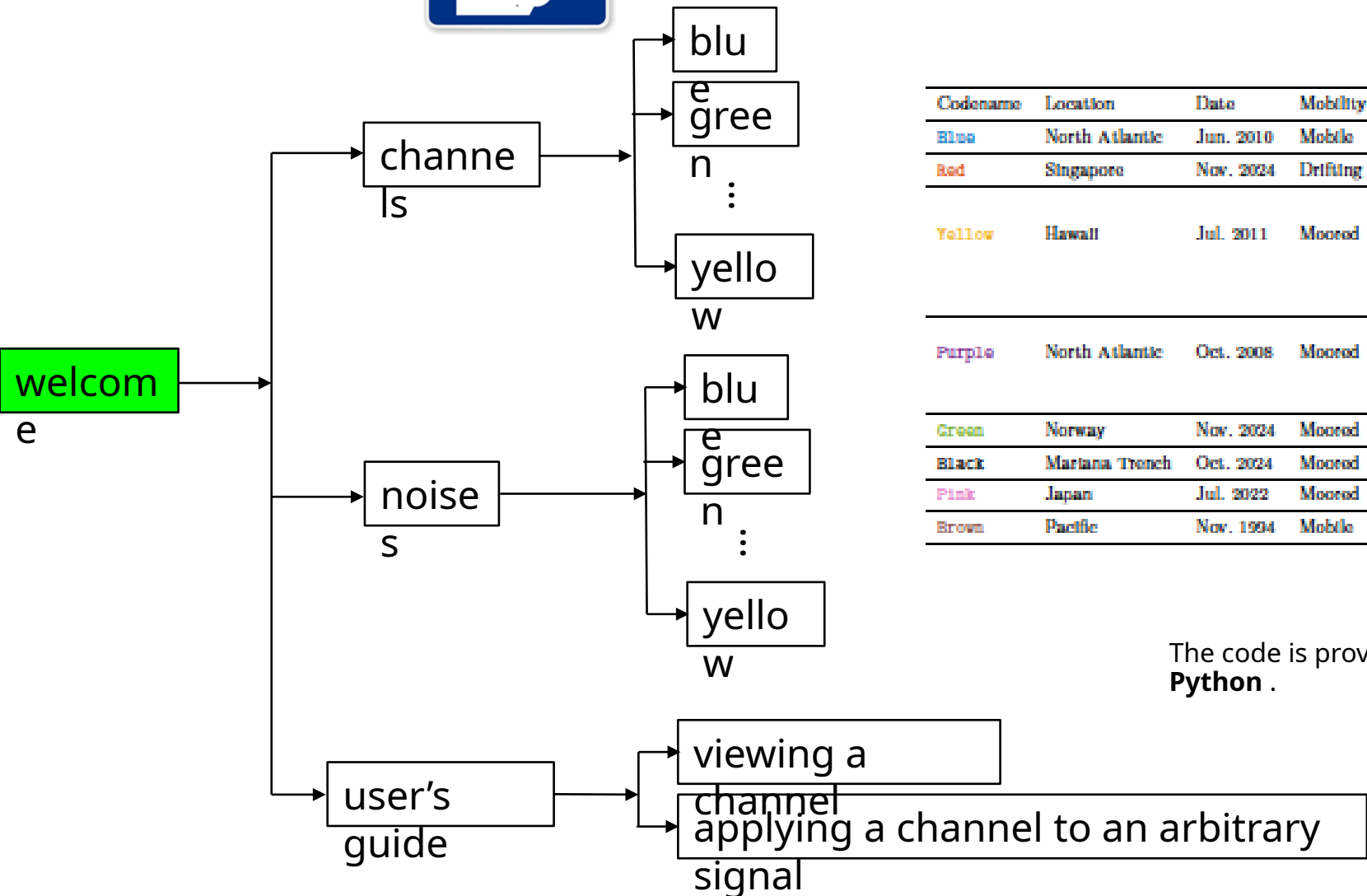
→ Blue
(Gaussian)

→ Red (alpha-stable)

Verification



Library



Codename	Location	Date	Mobility	$d_r/d_R/d_w$ [m]	d [km]	f_c [kHz]	R [ksym/s]	Array	M	ℓ [m]
Blue	North Atlantic	Jun. 2010	Mobile	30-60/50/100	3-7	13	$10^4/2048$	Vertical	12	0.12
Red	Singapore	Nov. 2024	Drifting	6/4.6/8-20	0.1-0.4	25	9.6	Vertical	3	0.8
Yellow	Hawaii	Jul. 2011	Moored	50/50/100	3	13	6.25	Vertical	24	0.05
					7				24	0.2
				50/8.6-65/100	3	16	3.75			
					7	16	3.75			
Purple	North Atlantic	Oct. 2008	Moored	11/10/15	0.06	12.5	$10^4/1536$	Cross	32	0.0375
					0.2			Vertical	24	0.05
					1			Vertical	12	0.12
Green	Norway	Nov. 2024	Moored	20/43/60	0.27	6	4.5	Time	64	-
Black	Mariana Trench	Oct. 2024	Moored	8718/6/8720	8.72	18	12.6	Planar	8	≥ 0.088
Pink	Japan	Jul. 2022	Moored	176/146/ x	14	6	4	Vertical	24	0.9-1.8
Brown	Pacific	Nov. 1994	Mobile	652/900-1600/ x	3250	0.075	0.0375	Vertical	20	35

The code is provided in **Matlab** and **Python** .

<https://uwa-channels.github.io/>

